

Design of a Real-Time Visual Analytics Decision Support Interface to Manage Air Traffic Complexity

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ABSTRACT

An essential task of an air traffic controller is to manage the traffic flow by predicting future trajectories. Complex traffic patterns are difficult to predict and manage and impose cognitive load on the air traffic controllers. In this work we present an interactive visual analytics interface which facilitates detection and resolution of complex traffic patterns for air traffic controllers. The interface supports air traffic controllers in detecting complex clusters of aircraft and further enables them to visualize and simultaneously compare how different re-routing strategies for each individual aircraft yield reduction of complexity in the entire sector for the next hour. The development of the concepts was supported by the domain-specific feedback we received from six fully licensed and operational air traffic controllers in an iterative design process over a period of 14 months.

Keywords: Visual analytics; Visualization design; Safety-critical systems

1 INTRODUCTION

Air Traffic Controllers (ATCos) perform a variety of complex and stressful tasks every day. Assuring safe separation between aircraft is perhaps the most crucial task in Air Traffic Control (ATC). However, ATCos should also be pro-active and avoid inefficiencies by examining developing situations in the airspace. In denser areas, where aircraft are converging, extrapolating a future situation is challenging, especially when aircraft speed variations are uncertain (e.g. due to a weather condition) [11]. Such traffic patterns are prone to cause conflicts and are considered complex and challenging. In complex traffic situations, the risk and difficulty of these tasks increase which can cause ATCos to reach their workload limit. This leads to the notion of sector saturation which occurs when ATCos are unable to accept additional aircraft entering their sector. In traditional air traffic management, a complexity manager is responsible for modifying flight plans within fixed sector boundaries to ensure that the traffic complexity remains within the safe workload limits for ATCos. However, advanced air traffic management concepts, such as dynamic sectorization [19], allow flexible sectors, which require management of traffic congestion independent of sector boundaries. This will change the traffic patterns and as a result, the mental models that ATCos had developed based on the similar patterns they often observed, will be seriously affected. As a result, ATCos' situation awareness [23] may decrease imposing high operational cognitive load on them. Such critical states call for design of new decision support tools which not only assist ATCos in quickly identifying and resolving conflicts but also to assist them in identification and resolution of complex traffic patterns that increase risk of potential conflicts in the near future.

In previous studies by the authors [28, 29], a visual analytics interface was developed which supported ATCos solely in their daily task of conflict detection and resolution. The interface improved

effectiveness in understanding the situations about conflicts for both ATCos [29] and novices [30].

To address the problem of early detection of emerging complex and unfamiliar traffic patterns, this work proposes a visual analytics interface that support ATCos mental models when complex traffic patterns, potentially leading to conflicts, emerge. The contribution of the proposed designs is as following. First, by analyzing geometrical properties of interaction between aircraft pairs (presented in Sec. 4), the tool detects and visualizes clusters of aircraft that create complex areas. Second, within each cluster unit, the extent to which aircraft state changes lead to complexity reduction is visualized and compared between aircraft in the cluster. Third, this work extends work on our previously designed composite glyph for conflict resolution [29]. In the current work, the glyph is further enhanced by integrating solutions to complexity reduction into solutions to conflict resolution.

2 RELATED WORK

This section reviews clustering techniques used for analyzing air traffic data both in the domain of aviation and visualization. We further review methods developed for analyzing air traffic complexity.

2.1 Clustering techniques for analyzing air traffic data

Clustering techniques are commonly used for visually analyzing various aspects of air traffic data. Density-based clustering has been used for learning variability of flight patterns [2, 6, 16] and weather impacts assessment on flight trajectories [4, 6]. Temporal clustering in combination with other filtering techniques [14] and spectral clustering [15] have been used to investigate procedural deviations from planned trajectory in terminal airspace. Hierarchical clustering has been used for identification of prevalent flight routes in the terminal [26] and the entire French airspace [12]. Spatio-temporal and dynamic similarity-based clustering has been used for characterizing flight trajectories [3, 25] and detecting optimized flight routes from the historical data [5, 24]. All reviewed techniques are restricted to analyzing large data sets offline while managing traffic complexity requires a real-time clustering technique which is sensitive to conflict prone situations.

2.2 Analyzing local complexity in air traffic

Traffic complexity is a measure which computes the complexity associated with a traffic situation solely depending on the geometry of trajectories [11]. Objects initial condition, the inter-dependency between conflicts, and aircraft speed uncertainties affect the level of traffic complexity [11]. In an attempt to address ATCos' workload, various approaches have been taken to describe traffic complexity. Laudeman et. al. [22] characterizes air traffic flow by calculating a weighted sum of several parameters (traffic density, number of heading, speed and altitude changes etc.) that exceed a specific limit. The approach has been extended by defining measures to describe other complexity aspects such as sector structure [18], aircraft flight attributes and potential conflicts between aircraft [7, 20, 21]. A common weakness of these metrics is that they do not consider the intrinsic effects of traffic disorder [10, 17] on the complexity [8]. To measure traffic disorder, two different approaches have been

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proposed in the literature. The first approach is based on a dynamic system modelling, where topological entropy is used as a metric to measure traffic disorder [8, 9]. The second, is based on geometrical properties of traffic complexity [1].

While a large body of work has been devoted to visualizing various aspects of flight trajectories and measuring the complexity caused by the traffic situation, a clear gap exists in the visualization of the emerging complex patterns and the presentation of possible complexity-reduction solutions to ATCos in real-time. To address this gap in this work we use the geometrical approach proposed by [1] as a base to compute traffic disorder and discover complex traffic patterns. Moreover, to further consider the effects of conflicts on traffic complexity (in addition to traffic disorder), the metric defined by [1] has been modified as described in Section 4.

3 DESIGN OBJECTIVE AND INTERFACE FUNCTIONS

A literature study on the state-of-the-art of traffic complexity measures complemented with operational information through a discussion session with an approach ATCo form our starting point. The questions asked were: What does ATC complexity mean to you? What information do you need to discover complex traffic patterns? Do you seek the information in a particular order? How do you handle complex traffic patterns once detected? The outcome of the state-of-the-art survey and the discussion session resulted in the specification of the following functions for the interface:

F1 Automatic identification of aircraft clusters that represent complex areas. Reading the information from the labels of aircraft can become time consuming, especially during congested situations. This interface function was derived to support ATCos in analyzing interaction between aircraft pairs and detecting areas of complexity.

F2 Visualization of cluster complexity level. When ATCos detect complex patterns, they prioritize resolving the one that looks most complex to them. This interface function aims to support ATCos in quickly identifying the most complex pattern.

F3 Highlighting of aircraft in conflict within a complex traffic pattern. When dealing with a detected complex pattern, the ATCos' first priority is to detect and resolve conflicts. This function was derived to keep ATCos aware of the situation about conflicts on top of managing the flow of the traffic.

F4 Real-time visualization of individual aircraft contribution to reduction of traffic complexity. To comply with airline demands, ATCos try to resolve complex traffic patterns by making the least number of changes. Re-routing an individual aircraft which contributes the most to reduction of complexity is therefore a preferred solution. Heading and speed variations among aircraft in a cluster are the two most contributing factors to traffic disorder [10]. Therefore, aircraft contribution to reduction of complexity was visualized in terms of both heading and speed variations.

F5 Visualizing effects of various resolution strategies on complexity reduction for each individual aircraft. When re-routing an aircraft, ATCos consider also heading information of the surrounding aircraft to make sure their selected strategy will not add to the current complexity. The interface function is derived to support them in visualizing the consequences of their strategy.

4 CLUSTER DETECTION AND COMPLEXITY ESTIMATION

This section describes how complexity of a region is measured independent of sector geometry, by identifying aircraft clusters (that contribute to traffic disorder) in three dimensions. Aigoin [1] proposes a single metric for measuring region complexity, by utilizing two geometric properties of interaction between aircraft, namely proximity and convergence. Proximity measures aircraft aggregation level, i.e., the maneuvering room available to the controller. Convergence measures the pace at which the aggregation level escalates. Proximity between aircraft i and j is given by $a_{ij} = e^{-\frac{1}{2}(\alpha_p d_{ij})^2}$

where $\alpha_p = 0.25$ and $d_{ij} = \sqrt{\frac{(x_i - x_j)^2 + (y_i - y_j)^2}{h_{sep}^2} + \frac{(z_i - z_j)^2}{v_{sep}^2}}$.

h_{sep} is the minimum horizontal separation equal to 5 nautical miles, and v_{sep} is the minimum vertical separation equal to 1000 feet. Convergence between aircraft i and j is given by equation 1,

$$b_{ij} = \begin{cases} -\frac{\partial(d_{ij})}{\partial t} & \text{if } \frac{\partial(d_{ij})}{\partial t} \leq 0 \quad (\text{convergence}) \\ 0 & \text{otherwise} \quad (\text{divergence}) \end{cases} \quad (1)$$

One limitation of the metric defined in the work of Aigoin [1] is that a converging pair of aircraft that are not mutually in a conflict, but relatively close is considered more complex than a converging pair that is in conflict but not closer than the former pair at the instant they are detected. According to the outcome of the conducted discussion session (Sec. 3), ATCos' first priority in dealing with complex patterns was to resolve conflicts. Therefore, in addition to the measures defined above, in this work we introduce a new measure as given by equation 2,

$$s_{ij} = \begin{cases} \frac{\sqrt{(h_{sep}^2 + v_{sep}^2)}}{d_{CPA}} & \text{if } i \text{ and } j \text{ are in a conflict} \\ 1 & \text{otherwise} \end{cases} \quad (2)$$

where d_{CPA} is the pairwise distance between aircraft (in a conflict) at the closest point of approach. Since aircraft in conflict violate the separation criteria, s_{ij} is always larger than 1. The smaller the value of d_{CPA} , the larger the value of s_{ij} . The complexity (c_{ij}) is calculated for each aircraft pair considering all three measures as $c_{ij} = a_{ij} \times b_{ij} \times s_{ij}$. This results in a single complexity matrix which quantifies the interaction of each aircraft pair in the whole airspace (equation 3). This matrix is then partitioned into disjoint clusters (sub-matrices) following a friends-of-friends (agglomerative) algorithm. In other words, for each aircraft pair, if the value of c_{ij} exceeds a threshold, the pair is considered as belonging to the same cluster. Clusters with common members are then iteratively merged until the partitioning of the matrix is complete.

$$C = \begin{bmatrix} 0 & c_{1,2} & \dots & c_{1,n-1} & c_{1,n} \\ c_{2,1} & 0 & \dots & c_{2,n-1} & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{n-1,1} & c_{n-1,2} & \dots & 0 & c_{n-1,n} \\ c_{n,1} & c_{n,2} & \dots & c_{n,n-1} & 0 \end{bmatrix} \quad (3)$$

The complexity value of each cluster is calculated using $C_l = \sum c_{ij}$ for every i, j that are members of the cluster l .

5 VISUALIZATION DESIGN PROCESS

The design process underlying the final version of the complexity visualization interface presented in this paper has spanned iterative development over 14 months. The prototype consisted of a real-time cluster visualization unit (Sec. 5.1) and a composite glyph-based visualization unit (Sec. 5.2). A user study was conducted with with 3 operational approach ATCos as presented in Sec. 5.3. Based on the outcome of the study, two domain experts with more than twenty years of experience as en-route ATCos were consulted which led to further enhancements of the prototype (Sec. 5.4).

5.1 Cluster visualization

The interface layout comprises of two main views: the situation display and the complexity visualization display as shown in Figure 1. The latter depicts aircraft forming a complex cluster on a circle (F1), with the circle's radius indicating the complexity level. Each circle's placement corresponds to the geometric centre of the aircraft on the situation display (F2). Within these circles, two spider graphs are shown representing each aircraft speed and heading variability relative to the other aircraft in the cluster. Appendix A demonstrates how these values are calculated. To visually represent these values they are normalized (divided by the average for all aircraft in the

cluster) and then mapped to the cluster circle (multiplied by the cluster radius). An aircraft with the largest relative heading (or speed) variability creates larger values in the complexity matrix defined in Sec. 4 and thus is contributing to the cluster complexity more than other aircraft. Therefore, changing the heading or speed of that aircraft will potentially lead to more reduction in the complexity of the cluster. (F4) (Fig. 1, right). The motivation for the choice of a spider graph was to communicate the information about bivariate data (speed and heading contribution to the reduction of complexity) effectively [27] without requiring the ATCo to look at two different views. The cyan graph represents heading contribution whereas the red graph represents speed contribution. On the spider graph, aircraft in a cluster are separated with equal distances. Since in the polyline-based visualizations patterns can be best interpreted between neighbouring axes [27], to direct ATCos focus towards the aircraft with larger effect on the complexity of clusters, aircraft position on the *cluster circle* are ordered by their contribution to the complexity of the cluster. This way, the top contributing aircraft become neighbours and as a result the non-overlapping surface area of the two polygons are maximized. To enhance the readability of the plot when several aircraft belong to one cluster the axes are separated with equal angles. Upon double clicking anywhere in the cluster, two functions are activated. First, the cluster and its associated aircraft on the situation display are emphasized by being highlighted in blue. Second, to facilitate identification of aircraft in the cluster, labels are assigned to them in both views arranged in alphabetical order. To further emphasize the aircraft in conflict within a complex cluster (F3), a box is placed around their labels. Figure 1 (right), shows A and D are in conflict with each other. The cluster visualization display supports ATCos in their decision making processes by filtering out the information in two hierarchical levels. First, the aircraft that seem to be close in 2D on the situation display but far vertically, as well as the aircraft which are close but will soon diverge are filtered out (green aircraft shown in Fig. 1(left)). Second, the spider graph visualization enables the ATCO to quickly identify which aircraft require speed or heading change, allowing for an efficient reduction of cluster complexity with minimal changes to aircraft states. Upon clicking on an aircraft icon on the cluster, a glyph appears as depicted in Figure 2. Figure 2 depicts the glyph consisting of only the visual encodings which describe solutions to conflicts (not complexity management). This glyph was designed in our previous work [29]. The visual encodings which describe solutions to manage traffic complexity are explained in Section 5.2.

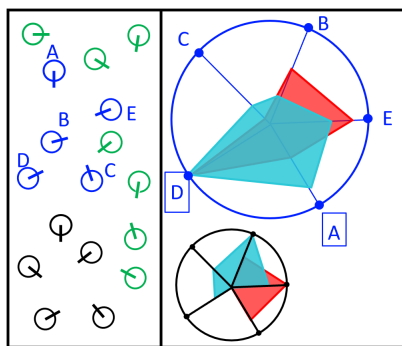


Figure 1: The interface consists of a situation display (left), where the entire traffic is shown and a complexity visualization (right), where two complex clusters are visualized. The selected cluster is colored in blue with its members highlighted and labeled on both displays. A box around labels indicate aircraft A and D are in conflict. The cyan and red spider graphs represent aircraft contribution to the complexity wrt. heading and speed variability. For clarity in the figure, the aircraft not belonging to any clusters are colored in green and those belonging to the less complex cluster are colored in black.

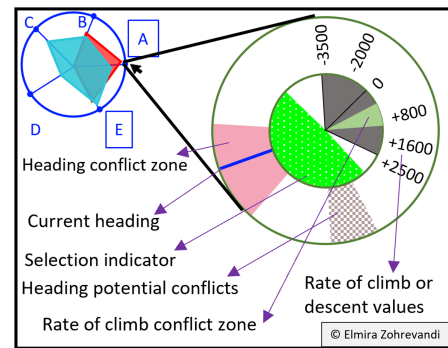


Figure 2: Selecting an aircraft in the cluster (aircraft A), displays the composite glyph. For clarity, real scales for cluster size and glyph size are ignored in this figure. The figure only depicts the visual encodings for solutions to conflicts as developed and described in detail in [29].

5.2 Glyph visualization

A composite glyph was composed to integrate the new visual encoding designs for complexity management into the previously designed visual encodings for conflict resolution. Figure 3 depicts the integrated glyph. The contribution of the current study to this composite glyph is thus the addition of potential solutions to traffic complexity which are shown by two plots. The blue graph shows the cluster complexity fluctuations across a 80-degree heading range, centered on the current heading. The red graph shows the cluster complexity fluctuations across a 40-knot speed range, centered on the current speed. This visualization is contained within a rectangular frame, referred to as the speed plot box (Fig 3). This box is consistently oriented parallel to the heading direction, with the current speed indicated by a red dashed arrow. Within the confines of the speed plot box, numerical speed change values are displayed. The positive direction of the speed axis is towards the heading direction. This orientation design for the speed plot is expected to be intuitive as it complies with reality where the aircraft velocity vector is always aligned with the track. In the initial design, the heading plot was also enclosed inside a rectangle. However, obtaining the complexity for heading values near the box corners was difficult. Therefore, the heading plot was projected onto an arc section, attached to the glyph's outer circle. The plot is drawn in polar coordinates, where the angular and radial coordinates correspond to heading and complexity values, respectively.

The newly designed glyph gives additional insight about the effect of various strategies on cluster complexity (F5). For example the heading plot visualization in Figure 3 shows that decreasing the current heading to *heading change (option 1)* in the figure, will resolve the conflict but will increase the cluster complexity. It is also a risky choice as the selected heading is close to the heading zone for potential conflicts (depicted as patterned section). However, increasing the heading to *heading change (option 2)* in the figure, will both resolve the conflict and decrease the cluster complexity.

5.3 User study

A qualitative user study (see Appendix B for details) was performed to investigate the designed concepts usefulness for ATCos. Three fully licensed approach ATCos with an average of 11 years of experience participated in the study. The experiment consisted of three steps; a training session, a simulation session followed by a semi-structured debriefing. (A) First a 10-minute training video introduced the designed concepts and interactive features. (B) Then ATCos worked with the tool on the designed scenario. Their task was to report the number of complex patterns they detect in the traffic, sort them based on the level of complexity they perceive and report which aircraft state they would change to resolve the

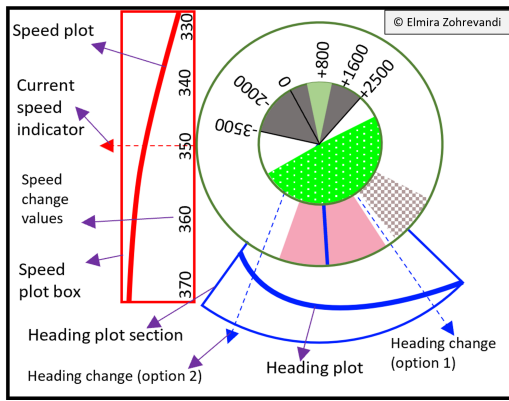


Figure 3: The designed glyph with complexity resolutions integrated with the conflict resolution glyph. The blue and red plots depict cluster complexity variations with heading and speed changes for the selected aircraft. To facilitate visual comparison between the two strategies, for each glyph, cluster complexity is normalized between (0, 1).

complexity and how. Interaction with the tool was done by means of keyboard and mouse. The three main independent variables were cluster complexity level, difficulty level in resolving the cluster complexity and display condition. Two display conditions were used: a control condition consisting of a conventional situation display and the conventional situation display plus the complexity visualization display. (C) Afterwards, a semi-structured debriefing was conducted. Without the visualization support, all ATCos missed the detection of one cluster but detected others successfully. When sorting based on complexity level, one ATCo reported the two most complex clusters in reverse order. All ATCos selected different aircraft for re-routing. With the visualization support, all ATCos agreed with the algorithm's clustering and that only changing one aircraft's heading would be sufficient. All ATCos confirmed that *"with the visualization support they reacted faster to complex clusters and ignored low complexity ones with higher confidence"* and found the glyph *"a bit complex in the beginning but learned how to work with it as they received additional explanations during the experiment"*.

5.4 Final Complexity Visualization Interface

To explore potential design improvements, we consulted two experienced ATCos (other than those who participated in the user study) who worked on complexity estimation of the en-route environment. Based on the outcome of the discussions (accounted for in Appendix C) an interactive plot was added to the complexity visualization display shown on a separate and movable view. Figure 4 depicts the complexity visualization display, which was ideated, designed, and implemented by the first author. The plot visualizes how the complexity changes in a given sector for the next forty minutes. The vertical red indicator enables the ATCo to visualize how complex clusters emerge in future. The buttons to the left of the plot allows adjustment of the complexity range that the ATCo wishes to visualize. As can be seen from the figure, when exploring various resolution strategies on the complexity glyph, the effect on the complexity of the entire sector is visualized via a dashed line.

6 EXPERT'S REVIEW

An expert review session (accounted for in Appendix D) was held where we asked an approach ATCo (the same expert mentioned in Sec. 3) about the potential benefits and weaknesses of the final version of our interface and the extent to which the complexity metric used in the algorithm was understandable. The expert found the spider plot visualization non-intuitive at the beginning but after working with the interface learned how to read the plot and could use it easily. They further found the complexity plot visualizing complex

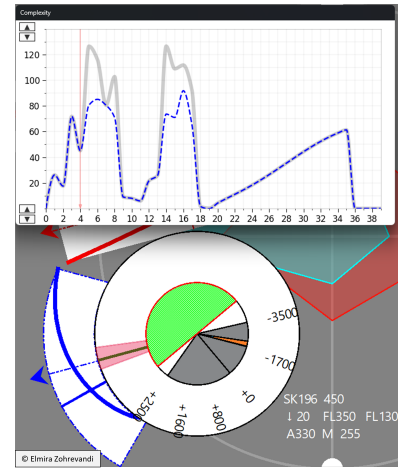


Figure 4: The finalized design of the complexity display, showing the interactive complexity estimation plot (solid grey line) with the consequences of a probed resolution strategy (dashed blue line) for an aircraft of a complex cluster (with spider plots in the background).

clusters of different time instances useful for the ATCo to form an understanding of complexity threshold of the algorithm and plan ahead when they should pay attention to the complexity display.

7 DISCUSSION AND FUTURE WORK

While our previous work focused on supporting ATCos solely in conflict detection and resolution tasks, the current work focused on creating insights into how the traffic situation evolves within the next hour. Despite the usefulness, our previously designed solution space glyph did not specify how different solutions affect the traffic situation as a whole. By visualizing complex traffic patterns, the encodings presented in the current work communicate the robustness of conflict-free solution strategies in real-time, enabling ATCos to consider more safety margins when possible. The advantage of doing this is to reduce the risk for conflict emergence. Even though ATCos found the glyph representation a bit complex, they could react faster and resolve complex patterns more efficiently with the visualization support. Given that it is a very steep learning curve to use new representations of complex information, the findings indicate that the proposed design can facilitate efficient management of complex traffic patterns for ATCos. Moreover, since traffic complexity drives ATCos' workload [13], our designed interface can support ATCos in managing their cognitive load. We intend to conduct an extensive evaluation study with more ATCos to validate both these aspects. In today's ATC, depending on the expertise that ATCos have gained over the years, they may have developed various definitions of complex traffic patterns as well as strategies for resolving them. An en-route ATCo often has a longer time to resolve conflicts than an approach ATCo, who deals with a much denser area where conflict situations may escalate rapidly. As a result an en-route ATCo may view heading variations as a complexity measure, while an approach ATCo may prioritize vertical separation. Future ATC concepts requires ATCos to perform various ATC strategies independent of the area they are controlling. By defining a unique term for complexity which encompasses a combination of complexity measures, our designed visual encodings can be a good candidate for training ATCos in detecting complex patterns in a consistent way regardless of the sector they are handling. We however, aim to fine tune the interface clustering algorithm by assigning weighting factors to complexity parameters depending on how traffic patterns are perceived complex among a larger population of ATCos. Beyond the ATC domain, another future direction is to generalize novel concepts to be used as a support for managing traffic complexity in other transportation domains (e.g. maritime and railway).

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